



Riding the Light

The tiny sail that could take us to the stars

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There's a romantic absurdity to the idea of a sailboat. You take a piece of cloth, catch some wind, and let the universe do the driving. It's a foundational technology that a Bronze Age fisherman left for us. Tucked inside that idea is something that physicists and aerospace engineers believe could one day carry a spacecraft to another star. Not in a thousand years or even a hundred, but in twenty.

However, here the "wind" is a laser the size of a small city. And the "sail" is thinner than a human hair. Welcome to the world of light sails – one of the most audacious, beautiful, and genuinely plausible ideas in the history of space exploration. And thanks to a team of researchers at Tuskegee University, that idea just got a whole lot more practical.

The problem with rockets

Before we get to the good stuff, a brief word about the vehicle that has carried every human being who has ever left the Earth: the rocket. I love rockets as much as the next person. They are extraordinary machines. They are also, when you step back and really think about it, a deeply inelegant solution to the problem of getting somewhere fast. To travel far, you need fuel. But fuel is heavy. Heavy things need more fuel to move. More fuel makes you heavier still. This vicious cycle, quantified in something called the Tsiolkovsky rocket equation which is the bane of every aerospace engineering student's existence, means there are physical limits on how fast and how far a rocket can take you.

Proxima Centauri, our nearest stellar neighbor, sits about 4.24

light-years away. At the speeds of our fastest current spacecraft, getting there would take somewhere in the neighborhood of 70,000 years. If your goal is interstellar exploration within a human lifetime, chemical rockets are, politely speaking, not going to cut it.

Enter the sail

Light sails work on a principle that sounds almost like a trick: photons, the massless particles that make up light, carry momentum. When they bounce off a reflective surface, they transfer a tiny push to whatever they hit. The pressure of sunlight on your outstretched palm amounts to the weight of a paperclip, but in the frictionless vacuum of space, tiny pushes add up. Keep pushing long enough, and you can go very fast.

The Breakthrough Starshot initiative, a project backed by the late Stephen Hawking and Russian billionaire Yuri Milner, proposed the most dramatic version of this idea: fire a powerful ground-based laser at a credit-card-sized probe equipped with a gossamer-thin sail, and accelerate it to 20% the speed of light. At that velocity, it would reach Proxima Centauri in just over 20 years. Not 20,000 years. Twenty.

Your sail shouldn't catch fire

The fact of the matter is that when you point an enormous laser at a thin piece of material, it tends to get hot. The whole point of a light sail is to reflect as much of the incoming laser energy as possible because energy you reflect is energy that pushes you forward, and energy you absorb is energy that turns your spacecraft into a tragic firework.

Previous light sail designs used metal-coated polymer films. These are reasonably reflective, strong, and manufacturable but they still absorb some of the laser energy.